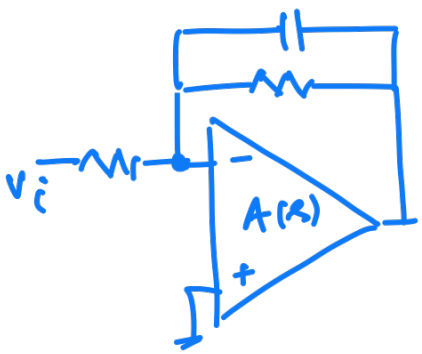
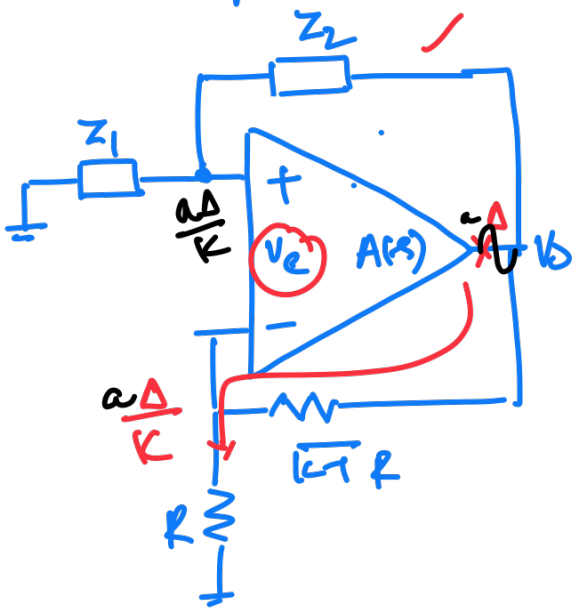
Lecture # 34



Wien-Bridge Oscillator.

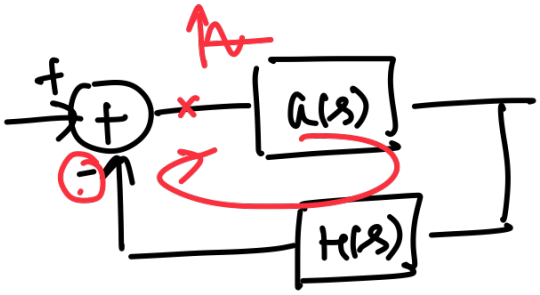


- $Z_1 = R_1$, $Z_2 = R_2$
- $\frac{V_+}{V_o} \cdot \frac{V_-}{V_o}$

$$\left|\frac{V_+}{V_o}(j\omega)\right| = \left|\frac{V_-}{V_o}\right| \quad \checkmark$$

$$\angle \frac{V_+}{V_o}(j\omega) = 0$$

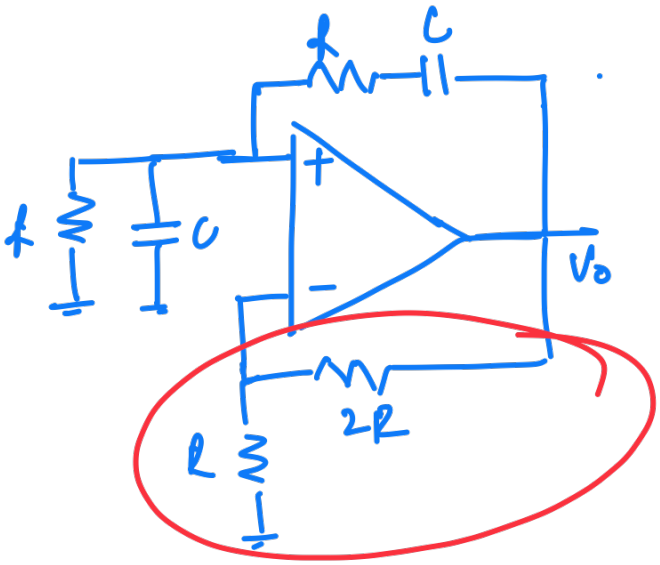
- $V_e = V_+ - V_-$



$$|A(j\omega_{osc})H(j\omega_{osc})| = 1$$

$$\angle A(j\omega_{osc})H(j\omega_{osc}) = \pi$$

$$LG = \left(\frac{Z_1}{Z_1+Z_2} - \frac{1}{K}\right) A(s)$$



$$V_- = \frac{V_o}{3}$$

$$V_+ = \frac{(R \,||\, 1/sC)}{\frac{R}{1+sRC} + R + \frac{1}{sC}} V_o$$

$$= \frac{sRC}{sRC + (1+sRC)^2} V_o$$

$$= \frac{sRC}{s^2R^2C^2 + 3sRC + 1} V_o$$

$$\frac{V_+}{V_o} = \frac{s/\omega_0}{\frac{s^2}{\omega_0^2} + \frac{3s}{\omega_0} + 1}$$

$$\left|\frac{V_+}{V_o}(j\omega)\right| = \left|\frac{V_-}{V_o}\right| = \frac{1}{3}$$

$$\left|\frac{V_+}{V_o}(j\omega_0)\right| = \frac{\left|j\omega_0/\omega_0 \quad 1\right|}{\sqrt{\left(1 - \frac{\omega_0^2}{\omega_0^2}\right)^2 + \left(\frac{3j\omega_0}{\omega_0}\right)^2}} = \frac{1}{3}$$

$$D(s) = \frac{s^2}{\omega_0^2} + \frac{3s}{\omega_0} + 1$$

$$\frac{s}{\omega_0} = \frac{-3 \pm \sqrt{9-4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$$

$$s = \frac{\omega_0}{2}\left(-3+\sqrt{5}\right), \frac{\omega_0}{2}\left(-3-\sqrt{5}\right)$$

$$\frac{V_+}{V_0} = \frac{s/\omega_0}{\left(\frac{s}{\omega_0} + \frac{3+\sqrt{5}}{2}\right)\left(\frac{s}{\omega_0} + \frac{3-\sqrt{5}}{2}\right)}$$

$$\angle \frac{V_+}{V_0}(j\omega_0) = 90^\circ - \tan^{-1}\left(\frac{1}{\frac{3+\sqrt{5}}{2}}\right) - \tan^{-1}\left(\frac{1}{\frac{3-\sqrt{5}}{2}}\right)$$

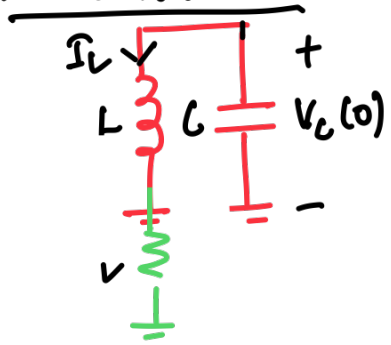
$$= 90^\circ - \tan^{-1}\left(\frac{2}{3+\sqrt{5}}\right) - \tan^{-1}\left(\frac{2}{3-\sqrt{5}}\right)$$

$$= 90^\circ - \tan^{-1}\left(\frac{\frac{2}{3+\sqrt{5}} + \frac{2}{3-\sqrt{5}}}{1 - \frac{4}{9-5}}\right)$$

$$= 90^\circ - 90^\circ = 0$$

$$L(s) = \underbrace{\left(\frac{Z_1}{Z_1+Z_2} - \frac{1}{k}\right)}_{\to 0} \frac{A_0}{(1+s/\omega_p)}$$

LC Oscillator



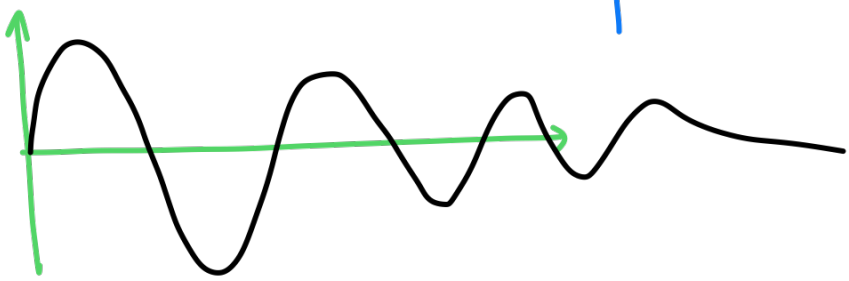
$$C\frac{dV_c(t)}{dt} = -I_L$$

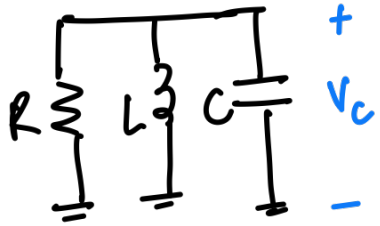
$$V_L = L\frac{dI_L}{dt}$$

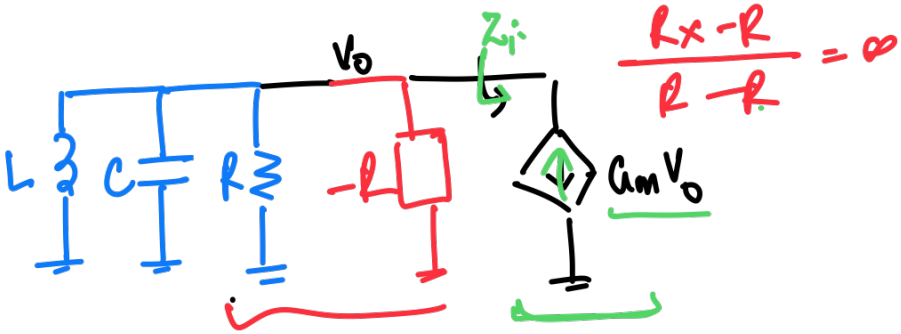
$$\frac{dV_L}{dt} = \frac{dV_c}{dt} = L\frac{d^2I_L}{dt}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$



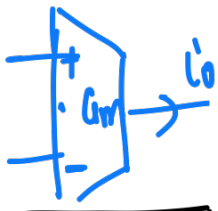




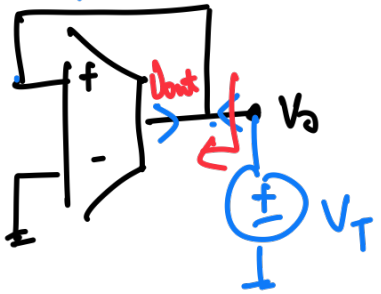
$$\frac{R_x - R}{R - R} = \infty$$

$$Z_i = \frac{-1}{G_m}$$

$$|Z_i| = R = \frac{1}{G_m}$$

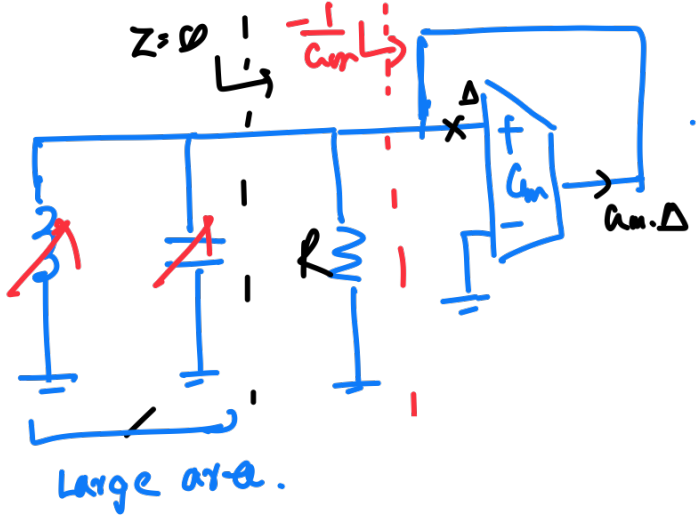


$$i_{out} = G_m V_c = G_m(V_+ - V_-)$$



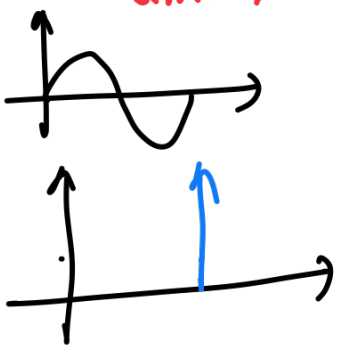
$$i_{out} = G_m V_T$$

$$Z = \frac{V_o}{-i_{out}} = \frac{-1}{G_m}$$



$G_m = \frac{1}{R}$

$G_m R > 1$



$$L_G(s) = G_m \left(R \,||\, \frac{1}{sC} \,||\, sL\right)$$

$$= G_m \frac{1}{\frac{1}{R} + sC + \frac{1}{sL}}$$

$$= G_m \frac{sLR}{s^2LCR + sL + R}$$

$$= G_m \frac{sL}{s^2LC + \frac{sL}{R} + 1}$$

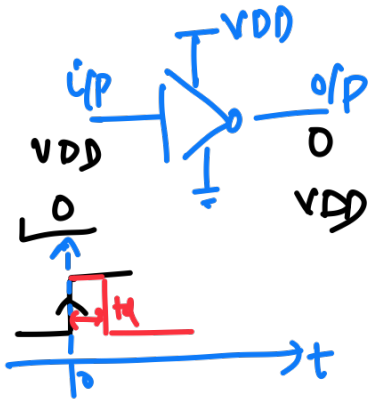
$$= \frac{sL/R}{s^2LC + \frac{sL}{R} + 1}$$

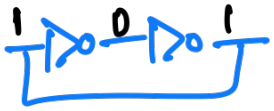
$$L_G(j\omega) = \frac{j\omega L/R}{(1 - \omega^2 LC) + j\frac{\omega L}{R}}$$

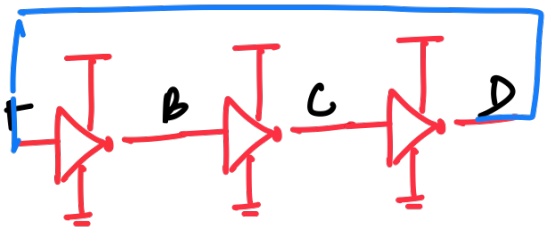
$$\angle L_G\left(j\frac{1}{\sqrt{LC}}\right) = 0$$

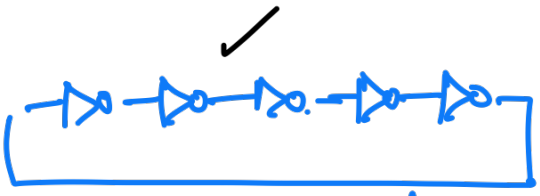
$$\left|L_G\left(j\frac{1}{\sqrt{LC}}\right)\right| = 1$$

Inverter based Oscillator.

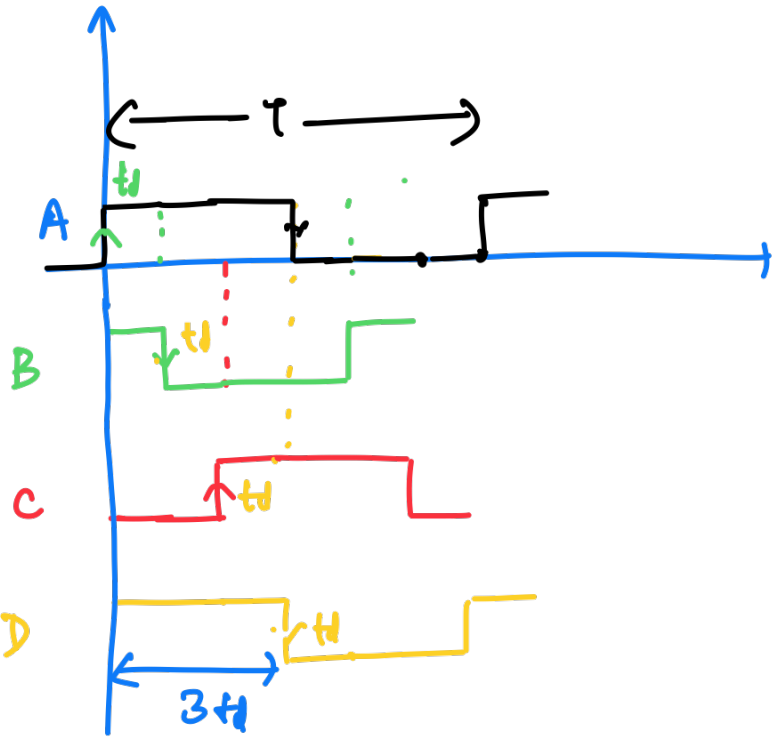








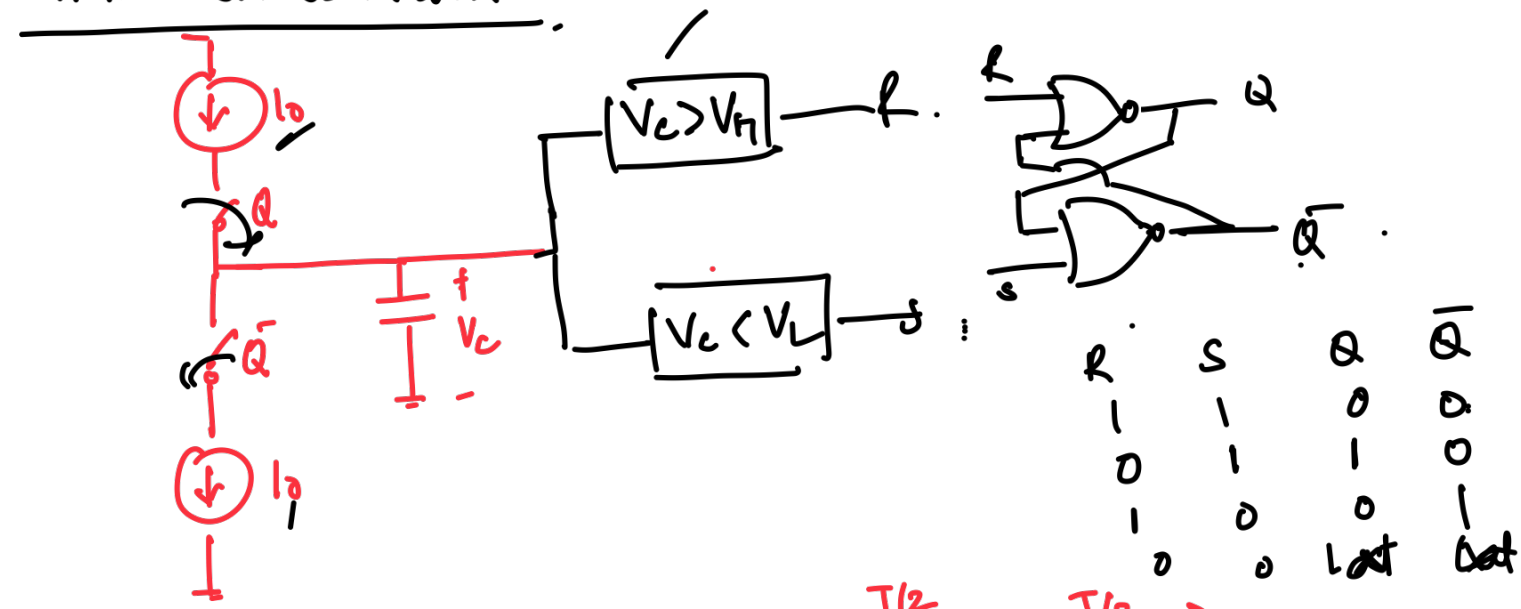
$T = \frac{1}{10 t_d}$



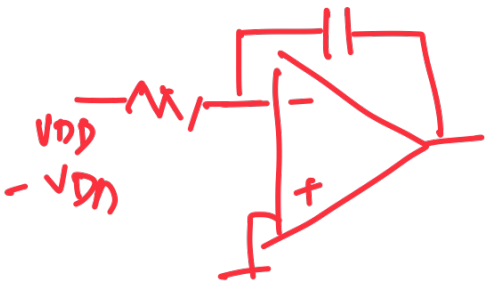
$T = 6 t_d$

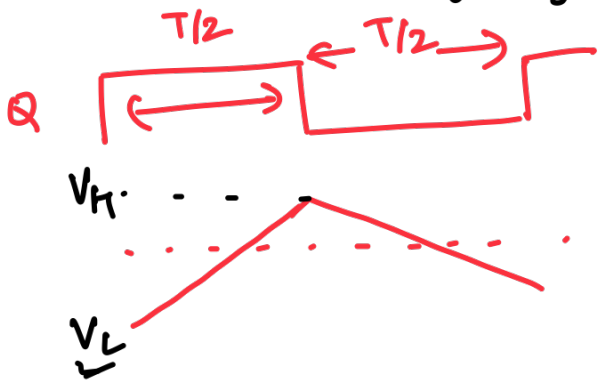
$t_d = f(V_{dd})$

Relaxation Oscillator.



R	S	Q	$\bar{Q}$
1	1	0	0
0	1	1	0
1	0	0	1
0	0	Last	Last





$$C(V_H - V_L) = I_0 \cdot \frac{T}{2}$$

$$T = \frac{2C(V_H - V_L)}{I_0}$$

